LICEO DE CAGAYAN UNIVERSITY
BACHELORS OF SCIENCE IN INFORMATION TECHNOLOGY

Subject: Application Development and Emerging Trends
Section: **BSIT - 2-2**
Group Name: **Group 12**
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1.1 Purpose

Campus Connect is to provide a centralized digital platform that streamlines student access to essential school information, resources, and extracurricular activities. By consolidating fragmented data into a single mobile-friendly interface, the application aims to enhance student engagement and simplify campus navigation. This document serves to formally **define all functional and technical requirements for the application**, establishing a comprehensive framework that ensures the final product meets the specific needs of the student body and administrative stakeholders.

1.2 Scope

Campus Connect is a streamlined information hub focused on public resource sharing and student privacy. The application features an **interactive map** with location filtering, a **building search tool**, and **live announcements** managed by authorized staff. To maintain security and ease of maintenance, the scope strictly excludes sensitive academic records, financial transactions, private messaging, and live GPS tracking.

1.3 Definitions and Acronyms

- **API (Application Programming Interface):** A set of protocols that allows **Campus Connect** to communicate with external services, such as Google Maps for the campus directory.
- **MVP (Minimum Viable Product):** The initial version of **Campus Connect** containing only essential features like the campus map to allow for early testing by students.

- **PWA (Progressive Web App):** The delivery method for **Campus Connect**, allowing students to access the app via a web browser and save it to their home screen without using an app store.
- **SDLC (Software Development Life Cycle):** The structured process followed to design, develop, and test the **Campus Connect** application from planning through deployment.
- **UI/UX (User Interface / User Experience): UI** refers to the visual layout and buttons students interact with, while **UX** focuses on the overall ease of navigation within the app.

2. Overall Description

2.1 Product Perspective

Campus Connect is designed as a **standalone product** that operates independently of the school's official administrative systems.

Unlike integrated student portals that handle sensitive data like grades or tuition, this app functions as a supplementary community layer. It acts as a "digital wrapper" for public campus information, existing as a self-contained web-based tool. While it may eventually use **APIs** to fetch public data, it does not require a direct connection to the school's internal database to function. This standalone approach ensures the app is easy to deploy, highly secure, and accessible to students without the need for complex official login integrations.

2.2 User Characteristics

Campus Connect serves three primary user groups with distinct needs. **Students** are the primary stakeholders; they are generally tech-savvy and require a fast, mobile-optimized interface with efficient search and filtering tools. **Administrative Staff** are the oversight group who may have varying technical proficiency and require a straightforward, secure dashboard to manage essential resources like cafeteria menus and official announcements.

2.3 General Constraints

Campus Connect must adhere to specific limitations to ensure a rapid, zero-cost deployment. The application **MUST** be built using a **No-Code or Low-Code framework** and hosted on a **free-tier cloud platform** to eliminate budget needs. The interface **MUST** be developed as a **Progressive Web App (PWA)** to ensure full accessibility on both **iOS and Android browsers**. The system **MUST NOT** store **Personally Identifiable Information (PII)** or sensitive academic data to ensure strict compliance with school privacy policies.

Part II: Detailed Functional Requirements

3. Functional Requirements

These define the specific features the app must provide to be useful on a daily basis.

ID	Feature Area	Description (Use SHALL)	Priority (H/M/L)
F-1.1	Interactive Resource Map	The system **SHALL** provide an interactive map with pins identifying key campus locations such as the library, gym, and student services.	High
F-1.2	Location Filtering	The system **SHALL** allow users to filter map pins by specific categories, including "Study Spots" or "Administrative Offices".	Medium
F-1.3	Building Search	The system **SHALL** allow users to search for specific campus buildings or services by name or keyword.	High
F-2.1	Live Announcements	The system **SHALL** feature a dedicated banner or scrollable section for real-time school-wide updates and event notices.	High
F-2.2	Announcement Management	The system **SHALL** allow authorized staff to update the announcement content via a simplified administrative interface.	High

4. Non-Functional Requirements

These define the quality and operational standards for the user experience.

Type	ID	Description (Use SHALL)
Performance	**P-1.1**	Every page navigation within the app **SHALL** load in under 2 seconds on a 4G network.
Security	**S-1.1**	Administrative access to the data spreadsheet **SHALL** be limited to authorized school emails.
Usability	**U-1.1**	The app **SHALL** be responsive and fully functional on both iOS (Safari) and Android (Chrome) browsers.
Availability	**A-1.1**	The application **SHALL** be accessible 24/7 with at least 99% uptime during the school semester.

Part III: Requirements Tracking and Verification

5. Requirements Traceability Matrix (RTM)

This table links the requirements to their testing status to ensure the app is ready for launch.

Req. ID	Description (Short)	Priority	Status	Test Case ID	Test Result
F-1.1	Search/Filter Clubs	High	In Progress	T-1.1	Pending
F-1.2	Resource Map Pins	High	Complete	T-1.2	**Pass**

P-1.1	< 2s Load Time	High	Testing	T-P-1.1	Pending
U-1.1	Browser Check	High	Complete	T-U-1.1	**Pass**

6. Conclusion and Future Work

Summary

The requirements for **Campus Connect** are now finalized, establishing a comprehensive framework for the application's initial release. This document clearly defines the functional features, technical constraints, and performance standards necessary to deliver a high-utility, standalone platform for the campus community. With the scope and verification methods established, the project is ready to move into the **development phase**.

Future Work

While the current version focuses on accessibility and core information sharing, several features were intentionally excluded to maintain a manageable initial scope. These may be integrated into a future version (Version 2.0) as the platform matures:

- **Push Notifications:** Implementing real-time alerts for urgent campus updates or reminders for club events.
- **Google Maps Integration:** Moving from a static or simplified interactive map to a full Google Maps API integration for real-time navigation.
- **Student Login Integration:** Future versions could explore secure Single Sign-On (SSO) with university credentials to allow for personalized schedules.
- **Event RSVP System:** Enabling students to "Check-in" or RSVP to club meetings directly within the app.
- **Dark Mode Support:** Enhancing the **UI/UX** by adding a dark-themed interface for improved nighttime readability.